

Deep Q-Learning for Spacecraft Docking under Clohessy–Wiltshire Dynamics

Jack Huston

Smead Department of Aerospace Engineering Sciences
University of Colorado Boulder

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Abstract

A Deep Q-Network (DQN) agent learns to soft-dock a chaser spacecraft with a passive target under Clohessy–Wiltshire (CW) relative orbital dynamics. The task is a finite-horizon discrete-action MDP in the target’s local-vertical / local-horizontal frame. The chaser fires either coarse or fine Gaussian noisy impulsive thrusts along a body axis. Five DQN seeds, each pre-trained from a buffer of heuristic-policy demonstrations, reach $99.2 \pm 1.2\%$ success at the proposed $0.5 \text{ m} / 0.05 \text{ m/s}$ capture box across 200 rollouts per seed with a mean terminal speed of 0.026 m/s . A theoretical ideal closed-form two-impulse CW transfer evaluated under the same thrust noise reaches the same capture box in all 500 rollouts, with roughly 23 times less Δv and 3 to 4 times more dock time. A 500-rollout Gaussian-ball perturbation study confirms 100% success within a 1 m scatter around the training nominal. Levels 1 and 2 of the original project proposal are both met. Level 3 is left for future work.

1 Introduction

Rendezvous and docking is a core guidance problem for orbital servicing, assembly, and crew-transport missions. Classical methods are model-based and rooted within analytical solutions. A typical guidance law solves a convex or nonlinear program, or in the simplest case inverts the state-transition matrix of a linearised relative-motion model to produce an impulse sequence that steers the chaser into a capture condition [1, 2]. The reinforcement learning approach used here takes a different route. The policy trains offline against a stochastic simulator, and at runtime it maps the relative state directly to a thrust command without solving an optimisation on the spacecraft.

This project trains a DQN [3, 4] policy for a soft-docking task against a Clohessy–Wiltshire [5] simulator and compares the learned policy to the closed-form two-impulse transfer [1] evaluated under the same stochastic dynamics. Although DQN methods have been used previously for this type of task, it serves as a demonstration for the capabilities of DQN methods applied to a representative rendezvous task and how the learned policy

compares to the fuel-optimal reference [1, 2].

The work follows the three success levels defined in the project proposal. Level 1 is a 2D minimum working example. Level 2 is a 3D main approach. Level 3 is an optional extension with more complex geometry or a different RL algorithm.

2 Background and Related Work

Clohessy–Wiltshire dynamics. For a chaser spacecraft in a near-circular low earth orbit at small relative distance from the target, the relative motion in the target’s LVLH frame obeys the linearised Hill–Clohessy–Wiltshire equations [1, 5]. These equations form an analytic 6-by-6 state-transition matrix $\Phi(\Delta t)$ parameterised by the orbital mean motion n .

Two-impulse transfer. The two impulse analytical solution [1, 2] partitions $\Phi(T)$ into position-position, position-velocity, velocity-position, and velocity-velocity 3×3 blocks to give a closed-form impulse pair. An impulse Δv_1 applied at $t = 0$ sets the chaser on a coast arc that reaches the origin at $t = T$, and an impulse Δv_2 at $t = T$ nulls the arrival velocity [1, 2]. The total Δv depends on the chosen transfer time, so a one-dimensional grid search over T gives the fuel-optimal transfer for a given initial state.

DQN and its stabilisations. DQN fits a neural network to the action-value function using off-policy Q-learning on a replay buffer with a slowly-updated target network [3]. Double DQN separates action selection and value estimation to reduce overestimation bias [4]. Both are used here.

Reward shaping. Dense shaping terms speed up learning but can move the optimum if the shaping is not potential-based [6]. The shaping used here is a sum of distance and velocity penalties, a state-dependent live cost, an approach corridor term, and terminal bonuses.

RL for spacecraft proximity operations. Prior work has applied RL to related problems, including meta-learned guidance for planetary landing [7],

cluttered-environment rendezvous with PPO [8], and proximity-operations guidance with DDPG [9]. Deep Q-learning from demonstrations [10] is the technique the 3D case borrows from to seed the replay buffer with roll-outs of a known-good policy. This project follows the same general process, but with a narrower scope.

3 Problem Formulation

The problem is modelled as a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, T, R, \gamma, H)$.

State. The MDP state consists of the chaser relative position and velocity in the target’s LVLH frame. In 2D, $s = (x, y, \dot{x}, \dot{y}) \in \mathbb{R}^4$. In 3D, $s = (x, y, z, \dot{x}, \dot{y}, \dot{z}) \in \mathbb{R}^6$, with x radial, y along-track, and z cross-track. Position components are divided by 15 m and velocity components by 0.5 m/s before the neural network input.

Action. The full action set includes discrete impulsive thrusts along one body axis at a time, plus a no-op (coast). Two magnitudes of impulse are available in the 3D model case. The coarse impulse has $\Delta v_c = 0.05$ m/s and the fine impulse has $\Delta v_f = 0.01$ m/s, for $|\mathcal{A}| = 13$ total in 3D ($6 + 6 + 1$). The original project only had 7 actions, but this proved to be too coarse for reliable capture in the 3D case. The fine magnitude thrusting allowed the 0.5 m / 0.05 m/s capture conditions to be reachable under the bounded execution noise. In 2D the action set with 5 coarse actions was used as an initial foundation. A larger action set makes discrete DQN harder to train, but was required to get successful captures with the target spacecraft.

Transition. Each step adds the commanded Δv (with Gaussian execution noise $\sigma = 0.05\Delta v$) to the velocity component, then propagates through one CW step with $\Delta t = 1$ s using the analytic state-transition matrix at mean motion $n = 1.13 \times 10^{-3}$ rad/s (400 km circular reference orbit). Execution noise is applied only when a non-zero impulse is commanded. The coast steps are purely deterministic and are defined by the CW propagation only. An out-of-bounds terminal (any position component above 60 m) keeps the episode cost bounded during exploration.

Reward. The per-step reward is a sum of shaping terms plus two terminal events,

$$r = -\alpha\|p\| - \beta\|v\| \mathbb{I}[\|p\| < r_b] - c_f\|\Delta v\| - c_c d_{\text{out}} - c_k \mathbb{I}[\text{KO}] - \tau(s).$$

Here p and v are the position and velocity sub-vectors, d_{out} is the out-of-corridor distance, and $\tau(s)$ is a state-dependent live cost. The live cost is zero when the chaser is inside the intended docking approach corridor and its

velocity points toward the port ($p \cdot v < 0$), and $\tau_0 = 0.1$ otherwise. This behavior allows the policy to coast along the natural CW arc, when the geometry permits it, without being penalized. The fuel term scales with the commanded impulse magnitude, so fine impulses cost proportionally less than coarse impulses. The terminal events are a capture bonus of $+R_{\text{succ}} = 300$ when all $|p_i| < 0.5$ m and $\|v\| < 0.05$ m/s, and a collision penalty of $-R_{\text{coll}} = 50$ when $\|p\| < 0.1$ m with $\|v\| \geq 0.1$ m/s, or when the chaser leaves the box $[-60, 60]^d$. The remaining coefficients are $\alpha = 0.1$, $\beta = 5$, $r_b = 2$ m, $c_f = 0.2$ per m/s of Δv , $c_c = 1$, and $c_k = 2$.

Horizon and discount. The episode horizon is $H = 200$ steps and the discount factor is $\gamma = 0.99$.

Geometry is shown in Fig. 1.

LVLH frame: corridor (wire), capture + keep-out volumes

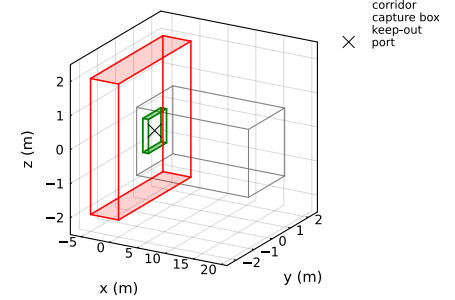


Figure 1: LVLH-frame layout. The gray wire-cage shows the approach corridor along $+x$. The green box is the 0.5 m capture volume at the origin. The translucent red box is the keep-out volume on the $-x$ side of the port.

4 Solution Approach

Q-network. A two-hidden-layer MLP with ReLU activations maps the normalised state to one scalar per action. The 2D network uses 64 units per layer and the 3D network uses 128.

Training. Experience tuples are written to a ring buffer of capacity 1.5×10^5 . Each environment step draws a minibatch of 128 and takes one gradient step of Huber-loss Q-learning with Adam at learning rate 10^{-4} . Action selection is ϵ -greedy. The bootstrap target uses Double Q-learning, in which the online network picks a^* and the target network evaluates it. The target network tracks the online network through a Polyak soft update $\theta_t \leftarrow (1 - \rho)\theta_t + \rho\theta$ at $\rho = 5 \times 10^{-3}$.

Best-policy snapshot. Even with Double DQN, Huber loss, and soft target updates, plain DQN on this task collapses mid-training. As often occurred, a policy that

reaches 90% success at one point can then later regress toward 0% a few hundred episodes later. To produce a stable deliverable, the greedy policy is evaluated every 100 episodes on a multi-cluster initial-condition bank of 8 representative starts with 5 jittered rollouts each, and the Q-network is snapshot whenever the pooled success rate improves on the best seen so far. Evaluations use that snapshot.

From-scratch versus off-the-shelf. The CW dynamics and state-transition matrix, the 2D and 3D environments, the reward and geometry code, the replay buffer, the DQN training loop with Double-DQN targets and Polyak updates, the two-impulse baseline, the Monte Carlo harness, and the figure generation were all written from scratch for this project. Flux.jl provides the neural-network layers and the Adam optimiser. Plots.jl, JLD2.jl, and CSV.jl handle plotting and IO. No off-the-shelf RL framework such as ReinforcementLearning.jl is used.

5 Analytical Baseline: Two-Impulse CW Transfer

A closed-form two-impulse transfer serves as the calibration reference for the DQN numbers. This partitions the 6-by-6 CW state-transition matrix for transfer time T into 3×3 blocks,

$$\Phi(T) = \begin{bmatrix} \Phi_{rr} & \Phi_{rv} \\ \Phi_{vr} & \Phi_{vv} \end{bmatrix}.$$

Choose Δv_1 so that the post-impulse coast arc arrives at the origin at $t = T$,

$$\Delta v_1 = -\Phi_{rv}^{-1} \Phi_{rr} r_0 - v_0.$$

The arrival velocity is $v(T^-) = \Phi_{vr} r_0 + \Phi_{vv} (v_0 + \Delta v_1)$, and the second impulse $\Delta v_2 = -v(T^-)$ nulls it. A one-dimensional grid search over $T \in \{40, 50, \dots, 200\}$ s returns the minimum- $\|\Delta v_1\| + \|\Delta v_2\|$ solution (minimum fuel usage) for a given initial state.

The baseline runs open-loop under the same thrust-noise model as the DQN, with execution noise applied at the two impulses only and deterministic coast in between. It draws initial conditions from the same distribution the DQN trains on to ensure a fair comparison.

6 Results

Summary. Table 1 gives the one-line status of each proposal level. Level 1 is met. Level 2 is met at the original proposal tolerance after adding a demonstration pre-training step. Level 3 was not attempted, and is listed under future work.

Level 1: 2D MWE. The 2D DQN converges within roughly 250 episodes. Across 100 held-out rollouts, the agent succeeds 100% of the time with a mean docking time of 47 s and a mean capture speed of 0.06 m/s. The 2D system, meant to test the foundational idea only, runs at a looser capture condition than the 3D case (1.0 m position and 0.15 m/s velocity). The 2D training curve, a representative trajectory, and the learned value function are in Fig. 2a. The agents’s intention to first maneuver into the approach corridor, then to approach the docking interface can be clearly seen in the run below.

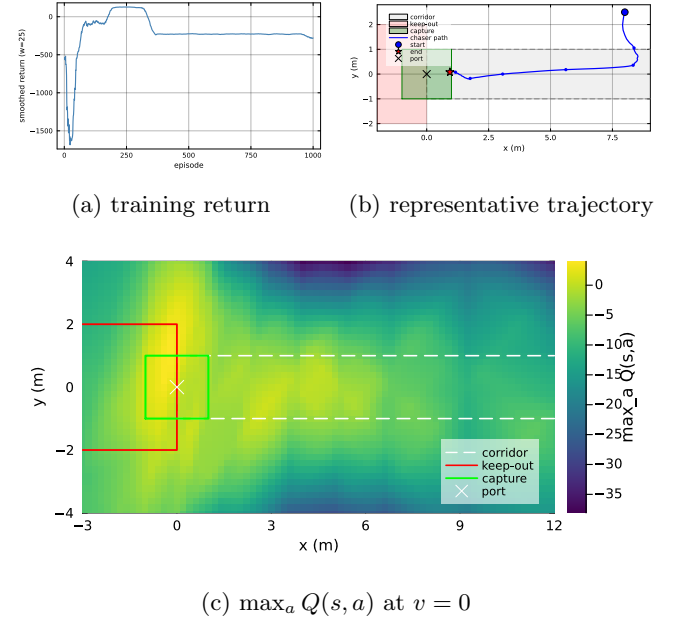


Figure 2: Level 1 (2D) results. (a) Training return rises from -1000 to positive by episode ~ 250 . (b) Representative 2D docking trajectory zoomed to path bounds. (c) Learned value function sliced at zero velocity.

Level 2 attempt 1: standard DQN. A first attempt at a 3D implementation trained five seeds of standard DQN directly against the proposal’s 0.5 m / 0.05 m/s capture condition. Every seed returned 0% success. Subsequent analysis shows that the failure is most likely due to difficulty exploring the state space as the agent almost never stumbles into a successful dock by random exploration. A hand-crafted heuristic policy was able to reach the capture condition in 100 of 100 rollouts on the same simulator. A random policy was able to reach the capture condition in about 1 of 1000 rollouts, or 0.1%. With the positive terminal reward that relatively rare, the system has nothing to propagate for thousands of gradient steps as it never reached the terminal reward, and the Q-function appeared to just settle into a hover-near-target equilibrium within the approach corridor instead of moving to the capture region. Training at a looser 1.0 m / 0.15 m/s capture condition trains

Table 1: Status of the three proposal-defined success levels.

Level	Task	Capture Gate	Status	Results
Level 1	2D MWE, 4-D state, 5 discrete actions, fixed corridor and keep-out	1.0 m / 0.15 m/s	met	100% success on 100 held-out rollouts
Level 2	3D main, 6-D state, 13 discrete actions (coarse + fine thrust), multiple ICs, corridor + keep-out	0.5 m / 0.05 m/s (original proposal)	met with demonstration pre-training	$99.2 \pm 1.2\%$ success, 0.026 m/s mean capture speed, 5 seeds
Level 3	Conical approach corridor, spherical keep-out, or PPO swap	—	not attempted	see Future Work

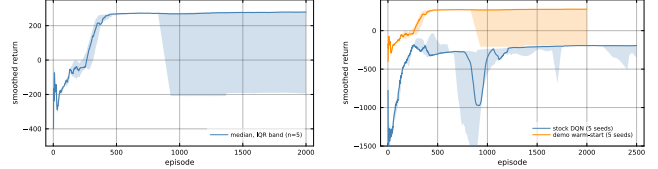
with better results (seed-average $58 \pm 20\%$ success) but the resulting policies do not transfer to the tighter capture tolerance. Mean terminal speed at convergence from that training sits around 0.08 m/s, just above the 0.05 m/s requirement.

Level 2 attempt 2: DQN with demonstration pre-training. The working level 2 system pre-fills the replay buffer with 400 rollouts of the hand-crafted heuristic that contain about 26 thousand "expert" transitions that cover the capture-approach region of the state space. The Q-network then runs a short supervised-style warm-up for 4000 gradient steps. The loss then sums standard TD-error on the buffer batch and a DQfD margin term [10], which pushes $Q(s, a_{\text{expert}})$ above every other $Q(s, a')$ by a fixed margin.

After this pre-training phase the greedy policy already reflects something like the expert policy instead of a random initialization. The online ϵ -greedy loop starts at $\epsilon = 0.3$ rather than 1.0 so that random exploration does not overwrite the expert-shaped Q-values before TD can reinforce them. With these changes all five seeds reach the proposal's capture condition.

Table 2 reports 200 held-out rollouts per seed at three capture tolerances. At the proposal's 0.5 m / 0.05 m/s tolerance, success is 0.992 ± 0.012 across five seeds, with mean capture speed 0.025 m/s. The multi-seed training curve is in Fig. 3a. Fig. 3b overlays the two training curves. The standard DQN training return stays around -200 . The demonstration-pre-trained DQN training return climbs past $+275$ within 400 episodes and holds there for the remainder of training. It is clear from these results that the demo warm-start approach yields far better results.

Representative trajectory. Fig. 4 shows a DQN docking rollout from $(7, -2, -1.5)$ m with a small inbound velocity, overlaid on the open-loop two-impulse baseline from the same initial condition and the same thrust-noise stream. Four views (isometric, top, side, front) make the 3D geometry legible. The chaser immediately moves to enter the corridor along $+x$, passes the keep-out volume on the $-x$ side of the port, and terminates inside the capture box. The starting distance of 7.4 m sits outside the training range of 1.5 to 3.5



(a) Demonstration pre-trained DQN (blue) vs. trained DQN, median and demo pre-trained DQN (orange). (b) Stock DQN (blue) vs. trained DQN, median and demo pre-trained DQN (orange).

Figure 3: 3D training returns across five seeds, smoothed with a 100-episode window. Median and 25–75 % interquartile band, which is more robust than mean $\pm$ std under the two seeds whose online returns regress late in training. Demonstration pre-training removes the exploration barrier.

m radial and demonstrates that the policy can handle starting conditions it never saw during training. The baseline path traces a single clean coast arc between its two impulses, but ignores the desire to enter within the approach corridor as expected. The DQN path follows a sequence of small corrections that bleed off velocity as it approaches the capture box.

The various views in Fig. 4 demonstrate clear behavior influenced by the reward shaping. The initial z offset of -1.5 m is only 0.5 m outside the corridor's cross-track half-width of 1 m, and in the xz view the DQN and the baseline paths agree closely because of this. However, the initial y offset of -2 m is a full meter outside the corridor's along-track half-width. In the xy view, the DQN path clearly leaves the baseline early and drives back toward $|y| < 1$ m to enter the approach corridor before it makes net progress along x to the capture box. The corridor penalty dominates when the chaser is well outside the corridor, so the learned policy enters the corridor first rather than take the fuel-optimal direct arc (baseline approach).

Comparison to the two-impulse baseline. Table 3 reports the same metrics for the two-impulse baseline evaluated on identical initial conditions and identical thrust noise. The baseline is fuel-optimal by construction for a fixed transfer time, so its mean Δv is a lower bound on what any feedback policy can reach on aver-

Table 2: 3D evaluation across five demonstration-pre-trained seeds, 200 rollouts per seed, reported at three capture tolerances. $\bar{p}$ and $\bar{v}$ are mean terminal position error and terminal speed.

seed	succ _{1.0/0.15}	succ _{0.7/0.10}	succ _{0.5/0.05}	$\bar{p}$ (m)	$\bar{v}$ (m/s)
0	1.000	1.000	1.000	0.559	0.022
1	1.000	1.000	1.000	0.529	0.024
2	1.000	0.990	0.985	0.572	0.029
3	1.000	1.000	1.000	0.520	0.025
4	0.995	0.980	0.975	0.546	0.029
mean $\pm$ std	0.999 $\pm$ 0.002	0.994 $\pm$ 0.009	0.992 $\pm$ 0.012	0.545 $\pm$ 0.021	0.026 $\pm$ 0.003

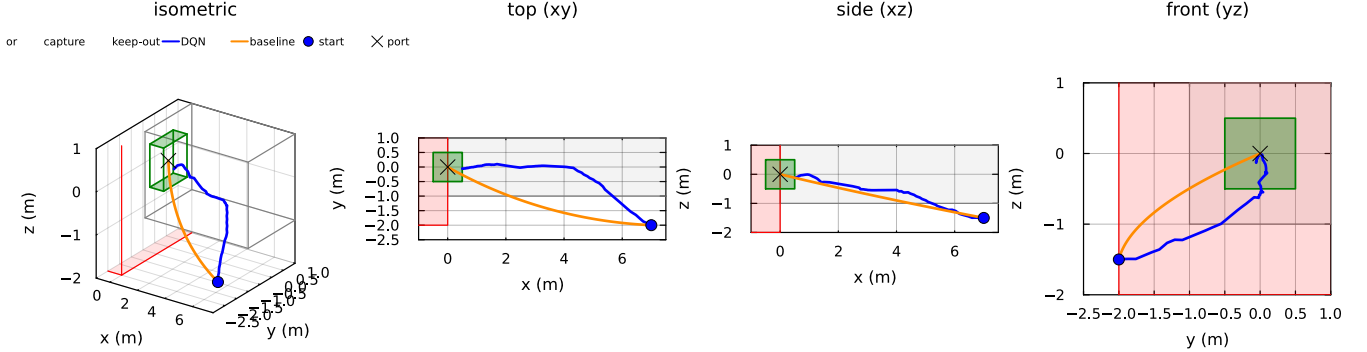


Figure 4: Representative 3D rollout from IC $\approx (7, -2, -1.5)$ m, outside the training range, shown in four views (isometric, top xy , side xz , front yz). The DQN path (blue) and the open-loop two-impulse baseline (orange) start from the same initial condition under the same thrust-noise stream. The wire-cage is the approach corridor, the green box is the 0.5 m capture volume, and the red box is the keep-out volume. Both paths stay clear of the keep-out and terminate inside the capture box. The baseline traces a single clean coast arc, and the DQN approaches through a sequence of smaller corrections.

Table 3: DQN vs. open-loop two-impulse CW baseline at the 0.5 m / 0.05 m/s capture condition, pooled over the same IC distribution and thrust noise, 100 rollouts per seed and 500 rollouts pooled. Δv is the total commanded impulse magnitude over the rollout.

method	succ	Δv (m/s)	t_{dock} (s)	$\ v\ _{\text{term}}$ (m/s)
DQN (5 seeds)	0.992	0.972	72.2	0.025
two-impulse	1.000	0.044	251.0	0.011

age. The baseline reaches the proposal capture condition in every one of 500 rollouts with mean Δv roughly 23 times smaller than the DQN. The DQN docks 3 to 4 times faster, although this comes at a heavy fuel cost.

Neither method dominates on both metrics. The baseline method, as expected, uses less fuel. The DQN performs better on docking time and works under closed-loop control. Under the assumption of Gaussian iid thrust noise used here, the DQN’s closed-loop structure does not show a visible accuracy gain over the open-loop baseline.

Monte Carlo perturbation study. A Monte Carlo study draws 500 initial conditions from

$\mathcal{N}(\text{nominal}, \text{diag}(\sigma_r^2 I, \sigma_v^2 I))$ with the nominal initial condition at $(2.5, 0, 0, 0, 0, 0)$, $\sigma_r = 1$ m per axis, $\sigma_v = 0.01$ m/s per axis, and evaluates seed 3 of the trained ensemble on each (arbitrarily chosen). Pooled success at the proposal capture condition is 100%, mean Δv is 0.80 m/s, and mean dock time is 62 s. Fig. 5 shows the trajectory bundle in the same four-view layout as the representative trajectory. The DQN’s desire to first maneuver into the approach corridor before approaching the capture box is clearly seen within the set of initial conditions. Additionally, it can be seen that all initial conditions successfully terminate within the capture box and avoid the keep-out zone. Fig. 5 shows the dock-time, action selection frequency, and total- Δv distributions.

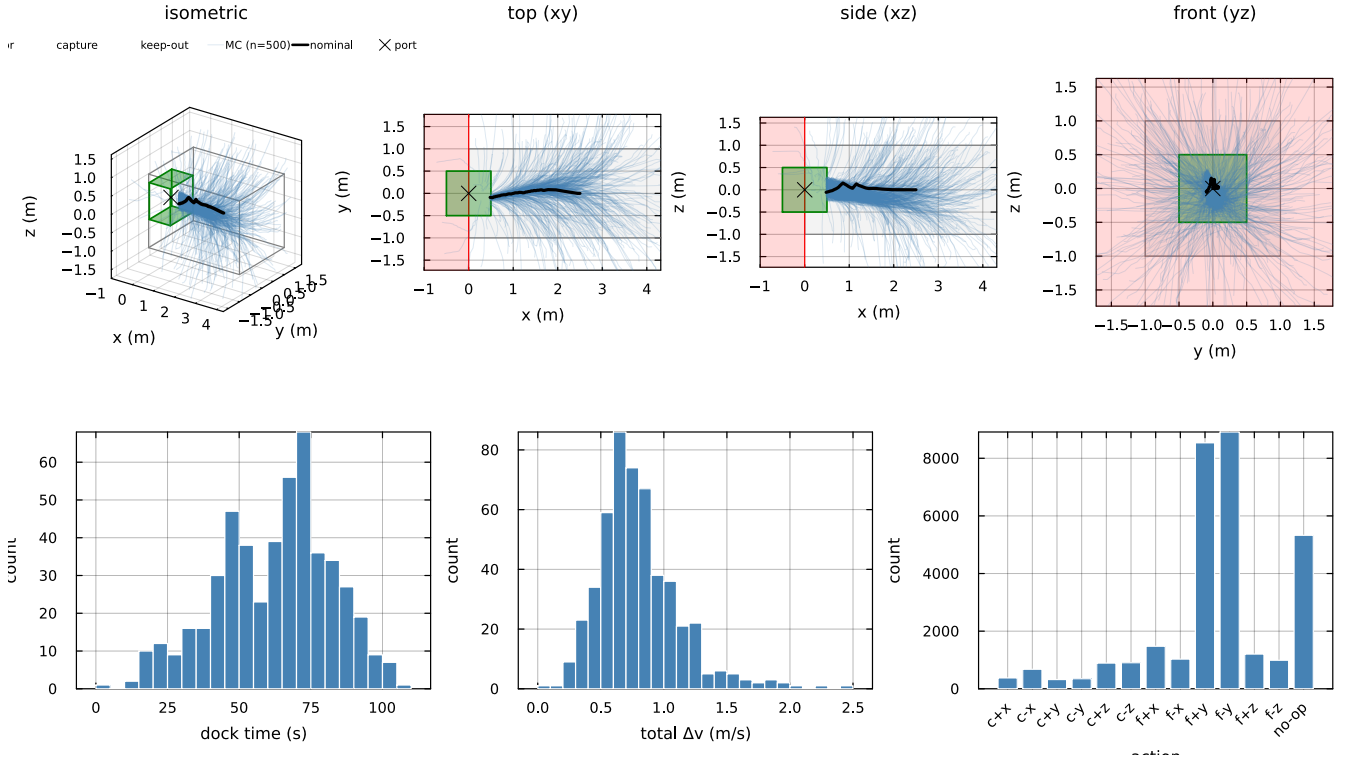


Figure 5: Monte Carlo perturbation study on seed 3, $n = 500$ draws from $\mathcal{N}(\text{nominal}, \text{diag}(\sigma_r^2 I, \sigma_v^2 I))$ with $\sigma_r = 1$ m and $\sigma_v = 0.01$ m/s around a nominal IC at (2.5, 0, 0) m. Top: trajectory bundle (blue) and nominal (black) in four views. Every path terminates in the capture box and none enters the keep-out. Bottom: dock-time and total- Δv distributions, plus action-usage histogram. The no-op dominates (coast phases), and the fine-magnitude actions (prefix f) dominate the terminal corrections near the capture box.

7 Conclusion

Standard DQN with the 13-action setup fails to reach the proposal’s 0.5 m / 0.05 m/s capture condition on any of five seeds. Random exploration had extreme difficulty finding the capture box, and was only able to enter it in fewer than 0.4% of rollouts and left the agent with no positive terminal reward to propagate. Seeding the replay buffer with 400 rollouts of a hand-crafted heuristic policy and adding a DQfD margin loss [10] to the pre-training phase removes that barrier. All five seeds then reach 97.5% or higher success at the proposal’s capture condition on 200 held-out rollouts, with pooled mean success 0.992 ± 0.012 and mean capture speed 0.026 m/s. The closed-form two-impulse CW transfer reaches the same capture condition 100% of the time with about 23 times less Δv and 3 to 4 times more dock time. The two methods optimize different points on the fuel-versus-time tradeoff under this noise model and could both be chosen depending on what optimum is required. A 500-rollout Gaussian-ball perturbation study confirms that the learned policy generalises uniformly within a 1 m scatter of the training nominal.

Status against the three proposal levels. Level 1 (2D Minimum Working Example) is met at 100% success on 100 held-out rollouts. Level 2 (3D Main Approach) is met at the original proposal tolerance of 0.5 m / 0.05 m/s with $99.2 \pm 1.2\%$ success across five seeds. Level 3 (conical corridor, spherical keep-out, or a PPO swap) was not attempted due to time constraints. The addition of the Level 3 goals would require its own training runs, seeds, comparisons, and figures. It is listed under future work.

Limitations. Results come from a bounded training distribution (1.5 to 3.5 m radial, ± 1.5 m lateral, ± 1.0 m cross-track). The agent has not been trained or tested against long-range starts or approaches significantly outside the geometry of the corridor. The thrust-execution noise model is Gaussian, zero-mean, uncorrelated step to step, and applied only when a non-zero impulse is commanded, which is optimistic relative to a real thruster. The baseline is open-loop, and a closed-loop MPC version would narrow the dock-time gap under noise and would be a fairer comparison between this approach and analytical close-loop guidance approaches. The final result heavily depends on demonstration pre-training, which requires a heuristic that already solves the task. The 3 to 5 times fuel gap against the baseline means the learned policy is not fuel-competitive with significant room for optimization.

Future work. The remaining gap to the baseline is primarily fuel used and not success rate. Three directions stand out in priority order. First, a fuel-weighted reward tune (raise c_f , add a per-episode Δv quadratic

term) to drive the learned policy toward the baseline’s fuel budget. Adjustments to the system weighting could have significant benefits in shaping the behavior of the resulting DQN agent. Second, potential-based shaping $F = \gamma\Phi(s') - \Phi(s)$ [6] to replace the current distance-penalty term so the optimum is not moved by shaping. Third, the Level 3 geometry extensions (conical approach corridor, spherical keep-out) and a swap to PPO. Policy-gradient methods like PPO tend to degrade less than DQN on shaped-reward tasks with larger discrete action sets and would likely return a better result.

Contributions and Release

This is a solo project. Jack Huston wrote the CW simulator, the 2D and 3D docking environments, the reward and geometry code, the DQN training loop with Double-DQN targets and Polyak soft updates, the multi-cluster best-policy snapshot selection, the DQfD margin behaviour-cloning loss, the two-impulse CW baseline, the Monte Carlo harness, the evaluation and figure pipeline, and the report itself. Claude Opus 4.7 with 1M context was used for the development of this project.

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From-scratch versus off-the-shelf. The following were written from scratch for this project: The CW dynamics and state-transition matrix, the 2D and 3D docking environments, the reward and geometry code, the replay buffer, the DQN training loop including Double-DQN targets and Polyak soft updates, the ϵ -greedy exploration schedule, the multi-cluster best-policy snapshot selection, the DQfD margin BC loss, the two-impulse CW baseline, and the Monte Carlo harness. Flux.jl provides the neural-network layers and the Adam optimiser. Plots.jl, JLD2.jl, and CSV.jl handle plotting and IO. No off-the-shelf RL framework such as ReinforcementLearning.jl is used.

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